

Spatiotemporal Variability of Modeled Watershed Scale Surface-Depression Storage and Runoff for the Conterminous United States

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Research Impact Statement: Correlation between simulated runoff and surface depression storage using continental-extent models to improve understanding and representation of surface-depression storage processes.

ABSTRACT: This study explores the viability of using simulated monthly runoff as a proxy for landscape-scale surface-depression storage processes simulated by the United States Geological Survey's National Hydrologic Model (NHM) infrastructure across the conterminous United States (CONUS). Two different temporal resolution model codes (daily and monthly) were run in the NHM with the same spatial discretization. Simulated values of daily surface-depression storage (treated as a decimal fraction of maximum volume) as computed by the daily Precipitation-Runoff Modeling System (NHM-PRMS) and normalized runoff (0 to 1) as computed by the Monthly Water Balance Model (NHM-MWBM) were aggregated to monthly and annual values for each hydrologic response unit (HRU) in the CONUS geospatial fabric (HRU; $n = 109,951$) and analyzed using Spearman's rank correlation test. Correlations between simulated runoff and surface-depression storage aggregated to monthly and annual values were compared to identify where which time scale had relatively higher correlation values across the CONUS. Results show Spearman's rank values >0.75 (highly correlated) for the monthly time scale in 28,279 HRUs (53.35%) compared to the annual time scale in 41,655 HRUs (78.58%). The geographic distribution of HRUs with highly correlated monthly values show areas where surface-depression storage features are known to be common (e.g., Prairie Pothole Region, Florida). **Editor's note:** This paper is part of the Featured Collection — The Emerging Science of Aquatic System Connectivity Parts I and II: Heather E. Golden, Charles R. Kane, C. Nathan Jones, Natalie G. Nelson, Tamara A. Newcomer-Johnson, and Lora L. Smith — Guest Editors. Other papers of the featured collection were published in Volumes 55(2) and 55(3) of the Journal. The complete collection can also be found as a virtual issue at [https://onlinelibrary.wiley.com/doi/toc/10.1111/\(ISSN\)1752-1688.aquatic-system-connectivity](https://onlinelibrary.wiley.com/doi/toc/10.1111/(ISSN)1752-1688.aquatic-system-connectivity).

(KEYWORDS: hydrology; modeling; surface-depression storage; PRMS; MWBM.)

INTRODUCTION

The spatiotemporal heterogeneity of the hydrogeographic features (e.g., ponds, detention ponds, prairie potholes, topographic low points, microtopography, ditches, etc.) that result in surface-depression storage makes generalization of hydrologic connectivity between stored water and runoff difficult (Rosenberry

and Winter 1997; Leibowitz and Vining 2003; U.S. Environmental Protection Agency 2015). Modeling and remote sensing techniques have been used to develop transferable metrics for surface-depression storage representation (Yang et al. 2010; Amoah et al. 2013; Evenson et al. 2015, 2016; Vanderhoof et al. 2017; Hay et al. 2018), though parameterization and calibration techniques have limited these applications to specific areas of interest. The United States

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Geological Survey's (USGS) National Hydrologic Model (NHM) (Regan et al. 2019, 2018) is a national-extent infrastructure which allows multiple deterministic hydrologic models at different temporal resolutions to be run on the same spatial units, referred to as hydrologic response units (HRUs). Previous studies have explored calibration techniques using proxies for surface-depression storage (James and Roulet 2007; Amoah et al. 2013; Hay et al. 2018). This study investigates the viability of using simulated monthly runoff as a proxy for depression storage in national-extent hydrologic models such as the NHM.

The NHM provides the consistent spatial infrastructure needed for a conterminous United States (CONUS)-extent analysis of spatiotemporal variability of surface-depression storage process representation (Regan and LaFontaine 2017; Regan et al. 2018). Conceptualization of surface-depression storage processes have been commonly described as a “fill-and-spill” mechanism: the filling of a network of topographic depressions on the land surface before spilling to create a hydrologic connection to runoff (Leibowitz and Vining 2003; Spence and Woo 2003; Spence 2007; Shaw et al. 2012). Rather than tracking the state and behavior of a very large number of discrete surface depressions, an adaptation to this conceptualization was made in the NHM-PRMS to create a single depression storage area within each HRU. The surface-depression storage capacity is the aggregate sum of all depressions within each HRU. This model conceptualization is termed “merged, fill, and spill” (Regan and LaFontaine 2017; Hay et al. 2018), where storage for all surface depressions within an HRU is simulated as increasing until a specified capacity is reached, at which time the excess volume is spilled out of the HRU. The aggregated area of surface depressions is described in the model as a fraction of the total HRU area to allow for scalable comparison across the CONUS. In addition, while runoff is generally conceptualized as precipitation minus actual evapotranspiration (ET) resulting in overland flow to a stream network; in this study, runoff is an aggregation of surface and subsurface flow leaving an HRU and used as a proxy for “wetness.”

The spatial and temporal resolution over which hydrologic processes occur is not consistent (Blöschl and Sivapalan 1995; Skøien et al. 2003; Tetzlaff et al. 2011) and identification of critical or dominant hydrologic processes may be dependent on the spatial and temporal scale of the model representation and evaluation. This study applied the NHM infrastructure with two different model time steps and results were compared for each HRU for two temporal aggregations (monthly and annual) to evaluate the spatial variability of correlation between daily surface-depression storage and runoff. Differences in correlation

between the monthly and annual results can be used to distinguish the dominant temporal resolution of surface-depression storage processes. The correlation between simulated water storage in surface depressions and simulated runoff for each HRU at monthly and annual time scales is used to quantify the responsiveness of landscape storage to aggregated surface and subsurface flow for the CONUS.

This study provides a modeling analysis demonstrating how surface-depression storage can be represented in national-extent watershed-scale hydrologic model applications at daily and monthly time scales. While simplification of physical processes through algorithms is generally required for application of national-extent watershed modeling, insight into where modeling and process resolution could be improved through modification of algorithms or changes to spatial or temporal resolution could help ongoing development toward accurately describing historic conditions and our ability to forecast future conditions. The challenge of representing surface-depression storage in national-extent physical hydrologic models requires innovative calibration techniques, such as using monthly runoff as a proxy for surface-depression storage. Improved representation and calibration of surface-depression storage in these national-extent models will in turn improve hydrologic simulation and forecasting of critical physical, chemical, and biological functions of wetland ecosystems (Cohen et al. 2016; Rains et al. 2016).

METHODS

This paper builds from many previous contributions to the literature, including the Precipitation-Runoff Modeling System (PRMS) and the Monthly Water Balance Model (MWB) development. These models and their application over the CONUS as presented are not novel contributions nor the focus of this work. Descriptions of these models and their configurations are briefly discussed for context in this paper, with many references for the interested reader. The novelty presented in this study is the correlation analysis of simulations from two models (PRMS and MWBM) within the NHM infrastructure.

This study applies the NHM infrastructure (Regan et al. 2018), which includes (1) physical hydrologic models with modular, process-based code (Markstrom et al. 2015; Bock et al. 2016); (2) spatial modeling units and default parameter values, called the Geospatial Fabric (Viger and Bock 2014; Driscoll et al. 2017), and (3) climate forcing datasets (Figure 1). The Geospatial Fabric of the NHM is

composed of 109,951 hydrologic response units (HRUs) for the CONUS. The NHM HRUs are derived from the National Hydrography Dataset, NHDPlusV1 (U.S. Environmental Protection Agency, U.S. Geological Survey 2005). Sizes of the HRUs are based on drainage density and range from 1-km² to over 67,000 km², with an average size of 74 km². Of note, fewer than half (53,007 of the 109,951) of the HRUs in the NHM Geospatial Fabric have surface-depression storage. The NHM infrastructure can be configured so that both monthly (Monthly Water Balance Model, NHM-MWBM) and daily (Precipitation-Runoff Modeling System, NHM-PRMS) physical models are run on the same HRUs (Markstrom et al. 2015; Driscoll et al. 2017; Regan et al. 2019, 2018). The nationally consistent parameter derivation and discretization within the NHM infrastructure (Regan et al. 2019, 2018) allows for comparison of hydrologic processes across space and time within the NHM-PRMS as well as between the daily time step NHM-PRMS and the monthly time step NHM-MWBM. Specifications for model configuration are described further in Regan et al. (2018). This study focuses on the comparison of simulated variables from NHM-PRMS (surface-depression storage, Driscoll et al. 2018) and NHM-MWBM (runoff, Bock et al. 2018) using the same spatial and temporal domain.

The PRMS is a daily time step, modular, deterministic, distributed-parameter, physical-process watershed modeling system used to simulate streamflow, surface-depression storage and other variables by

process algorithms that are based on a physical law or empirical relation with measured or estimated characteristics (Figure 2a) (Markstrom et al. 2015; Regan et al. 2019, 2018). Distributed parameter capabilities are provided by partitioning the CONUS into HRUs in which the components of flow: groundwater flow (groundwater reservoir), interflow (gravity reservoir), and surface runoff (impervious and capillary reservoirs) are computed in response to precipitation, air temperature, and land surface and subsurface dynamics (Figure 2a). Daily maximum and minimum temperature and precipitation data from the gridded Daymet version 2 dataset (Thornton et al. 2016) were used for the NHM-PRMS climate forcings. The gridded Daymet climate data were spatially aggregated to the NHM HRUs using an area-weighted averaging algorithm using the USGS Geo Data Portal (Blodgett et al. 2011). Most hydrologic processes in the PRMS are represented through algorithms applied at the HRU scale (Figure 2a). Other processes are represented at the model domain and stream network scales (Markstrom et al. 2015).

The MWBM is a modular hydrologic accounting system that computes monthly estimates of components of the water cycle using concepts of water supply and demand, snow accumulation and melt, and soil-moisture storage (Wolock and McCabe 1999; McCabe and Markstrom 2007) (Figure 2b). Bock et al. (2016) developed a parameter regionalization scheme for the MWBM that transferred parameter values from gaged to ungaged areas, resulting in a

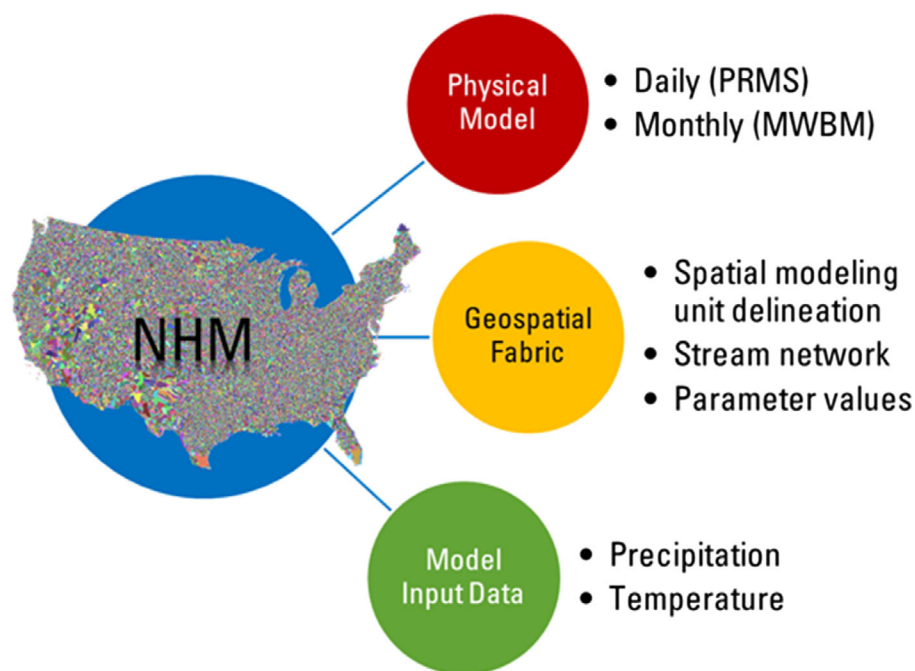


FIGURE 1. Schematic of the National Hydrologic Model (NHM) infrastructure and components: physical models (daily Precipitation-Runoff Modeling System [PRMS], and Monthly Water Balance Model [MWBM]), geospatial fabric, and model input data.

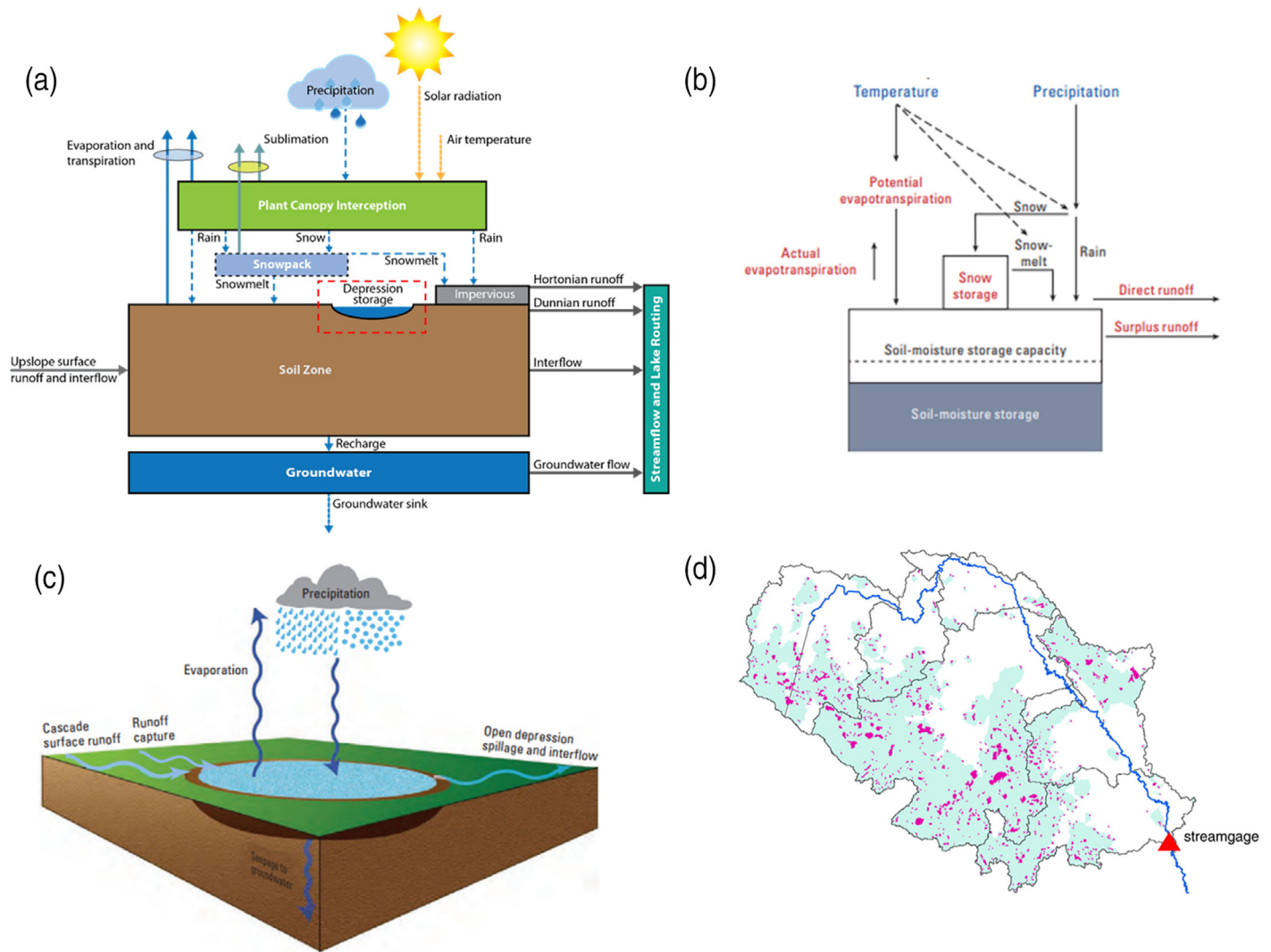


FIGURE 2. Schematic diagrams of (a) hydrologic processes and connections within the PRMS with the surface-depression storage process located in the dashed red box; (b) hydrologic processes and connection within the MWBM; (c) specifically the surface-depression storage process within the PRMS (modified from Regan and LaFontaine 2017); and (d) conceptual map modified from Hay et al. (2018) to show an example of hydrologic response units (HRUs) (grey lines), surface-water depression storage (magenta), contributing area to surface-water depressions (light blue area), and an outlet streamgage.

calibrated CONUS-wide application of the MWBM. Simulated values of runoff from MWBM applied by Bock et al. (2016) are used in this study.

Simulation of Surface-Depression Storage

The NHM-PRMS infrastructure (Regan et al. 2019, 2018) was used to simulate the effects of surface-depression storage on watershed response (Figure 2c) for the spatial extent of the CONUS. Previous simulations of surface-depression storage using the PRMS, have described applications in North Dakota (Vining 2002; Hay et al. 2018) and Georgia (LaFontaine et al. 2015). The surface-depression storage option within the Surface-Runoff Modules in the PRMS is described

in Markstrom et al. (2015) and Regan and LaFontaine (2017) and further for the NHM-PRMS in Regan et al. (2018). Techniques for deriving required input and parameter information, including the HRU delineation and attributes of surface depressions, is documented in Viger et al. (2010), and further in Regan et al. (2018).

Surface-depression storage area is characterized as the aggregate sum of NHDPlus water body features (e.g., ponds, detention ponds, prairie potholes, microtopography, ditches, etc.) within HRUs that are neither on-stream nor located where land cover data indicate impervious surfaces exist (Figure 2d). On-stream water bodies, such as lakes and reservoirs that are within or intersect with a 60-m buffer of NHM stream segments, are considered flow-through

features and are not included in surface-depression storage. This analysis relied on a 30-m digital elevation model associated with NHDPlus. Additional information on how surface-depression storage is defined within the model and methods for how parameters were derived can be found in Viger et al. (2010) and Regan et al. (2018).

The hydrologic process of surface-depression storage is conceptualized and simulated in the NHM-PRMS as a “merged, fill, and spill” (Hay et al. 2018). Conceptually, this allows representation at a landscape scale through merging all independent depressions within each HRU (Figure 2d) to allow for a single, lumped representation of the process for each HRU (Figure 2a and 2c). These conceptually merged depressions then allow water to move into (fill) and out of (spill), interacting with the larger hydrologic system (Figure 3).

The fill and spill movement of water into and out of the conceptually aggregated surface depressions in the PRMS is shown in more detail in Figure 3. Fractionally, aggregated surface-depression storage in each HRU can be designated as open (allowing spill) and closed (not allowing spill). For this study all aggregated surface-depression storage within an HRU was specified to be open. Water enters aggregated surface-depression storage in two ways: (1) direct precipitation and (2) overland flow captured by the calculated contributing area within the HRU. Water stored in aggregated surface depressions can leave the depression in three ways: (1) evaporation, (2) seepage to groundwater recharge, and (3) spillage, which is the sum of (a) interflow from storage above a threshold and (b) overland flow when storage reaches the maximum volume. In the current model representation, water cannot enter the aggregated surface depression through interflow or groundwater flow processes, though there is consideration for adding this capability to the code. Future reference to simulated surface-depression storage in this paper will be in this merged or aggregated sense; no individual surface depressions were modeled for this analysis.

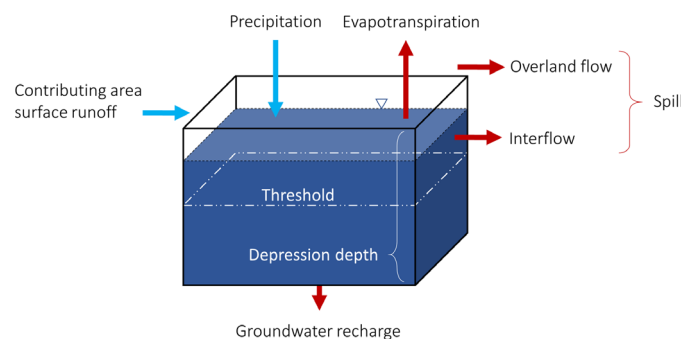


FIGURE 3. Schematic diagram the movement of water into (blue arrows) and out of (red arrows) surface depressions in the PRMS.

For convenience in calibration procedures, the surface area of aggregated surface-depression storage is specified as a decimal fraction (0–1) of the HRU area (Figure 4). The maximum fractional proportion of HRU area covered by surface depressions is represented by the PRMS parameter *dprst_frac* (Driscoll et al. 2017). The shape of the bottom of the surface depressions was specified to approximate a rectangular basin, though additional shapes could be specified (Regan and LaFontaine 2017). This consistent representation was chosen because there were not sufficient data to spatially distribute various shape parameters for each surface depression for the CONUS. The contributing area to the aggregated surface-depression storage is a separate parameter (*sro_to_dprst*) that is calculated using the underlying DEM for all NHM parameters.

Calibration

The NHM-PRMS was applied using a selected set of distributed parameters and calibration schemes (Driscoll et al. 2017; Regan et al. 2018). The Fourier Amplitude Sensitivity Test (FAST) (Cukier et al. 1975, 1973; Schaibly and Shuler 1973; Saltelli et al. 2006; Markstrom et al. 2016; Hay et al. 2018) parameter sensitivity analysis was used to identify the surface-depression storage parameters that had the most influence on surface-depression volume. Based on the FAST analysis, five PRMS surface-depression storage-related parameters, described in Table 1, were selected for inclusion in the calibration: average depth of storage depressions at maximum storage capacity (*dprst_depth_avg*), fraction of unsatisfied potential evapotranspiration (PET) to apply to surface-depression storage (*dprst_et_coef*), coefficient used in linear interflow equation for open surface depressions (*dprst_flow_coef*), coefficient used in linear seepage flow equation for open surface depressions (*dprst_seep_rate_open*), and fraction of open depression storage above which interflow occurs (*op_flow_thres*). Each HRU is assigned a unique value for each of these parameters from the model calibration.

A previous study indicated that a proxy for change in surface storage was needed to accurately parameterize aggregated surface-depression storage for the NHM-PRMS (Hay et al. 2018). In this study, normalized (0–1) NHM-MWBM runoff (Bock et al. 2016) was tested as a proxy for change in surface-depression storage. The parameters in Table 1 were calibrated using the Shuffled Complex Evolution global search algorithm (Duan et al. 1994, 1993, 1992), by comparing the normalized NHM-MWBM runoff (0–1) with the NHM-PRMS simulated fractional (0–1) surface-depression storage within each HRU. Two calibration

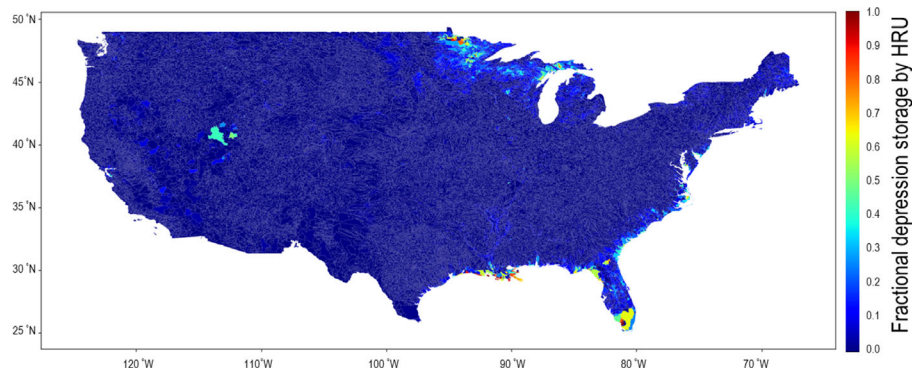


FIGURE 4. Spatial distribution of the fractional surface-depression storage (PRMS parameter “dprst_frac”) at the HRU resolution for the conterminous United States (CONUS).

TABLE 1. PRMS terms, descriptions and maximum, minimum, median, mean and standard deviation (StdDev) values for each used for calibration.

PRMS term	Description	Maximum	Minimum	Median	Mean	StdDev
dprst_depth_avg	Average depth of storage depressions at maximum storage capacity	10	300	49	80.4	79
dprst_et_coef	Fraction of unsatisfied PET to apply to surface-depression storage	0.75	1.25	1.04	1.02	0.2
dprst_flow_coef	Coefficient in linear interflow equation for open surface depressions	0.0001	0.1	0.048	0.049	0.029
dprst_seep_rate_open	Coefficient used in linear seepage flow equation for open surface depressions	0.00001	0.2	0.033	0.071	0.069
op_flow_thres	Fraction of open depression storage above which interflow occurs	0.75	1	0.83	0.85	0.07
scale_lower	Scaling factor for MWBM lower normalization limit	0.01	0.42	0.02	0.03	0.02
scale_upper	Scaling factor for MWBM upper normalization limit	0.51	1	0.91	0.87	0.14

factors (scale_lower and scale_upper; see Table 1) were introduced to set the lower and upper bound of the normalization for the NHM-MWBM runoff. This means that the normalized MWBM runoff value at 0 (or 1) could encompass more than just the very lowest (or highest) MWBM runoff values. The objective function (OF) to determine how close the simulated fractional surface-depression storage results are to “observed” NHM-MWBM normalized runoff values is calculated as follows in Equation (1):

$$\text{OF} = (\text{wgt1} \times \text{NRMSE}) + (\text{wgt2} \times (1.0 - \text{RS})) + (\text{wgt3} \times (1.0 - \text{RSann})), \quad (1)$$

where: OF is the sum of three OFs that calculate the (1) normalized root mean square error (NRMSE) between normalized NHM-MWBM runoff and NHM-PRMS fractional surface-depression storage on a monthly time step, (2) the Spearman’s rank correlation (RS) between normalized MWBM runoff and simulated fractional surface-depression on a monthly time step, and (3) the Spearman’s rank correlation

(RSann) between normalized MWBM runoff and simulated fractional surface-depression storage on an annual time step. The weighting of the OFs (wgt1 = 1.0, wgt2 = 0.5, and wgt3 = 0.25) were determined through trial and error.

The NHM-PRMS was run for a 30-year simulation time period, from January 1, 1980 to December 31, 2010. The first 10 years of this simulation time period were used as a spin-up period, meaning this period was used to initialize model states, and output from this time period were not used for further analysis. Results from the 20 years of this simulation, January 1, 1990 to December 31, 2010 were used in the calibration. Of the 109,951 HRUs in the NHM-PRMS, 56,944 were removed from the analysis because there was no surface-depression storage, leaving simulation of 53,007 HRUs with surface depressions for analysis.

The daily time series of output values for fractional surface-depression storage from the calibrated NHM-PRMS were aggregated and correlated with normalized runoff from the NHM-MWBM at monthly ($n = 240$) and annual ($n = 20$) time steps for each HRU with surface-depression storage. The Spearman’s rank correlation between NHM-PRMS

fractional surface-depression storage and normalized runoff from the NHM-MWBM was calculated by HRU and compared. This nonparametric and nonmonotonic correlation metric was used to test whether water storage in surface depressions leads to runoff at the HRU resolution and, thereby, infer an aggregated hydrologic connection to the stream network. Monthly correlation values allow for quantification of relatively short-term (intra-annual) time scale, whereas annual correlation allows for quantification of relatively long-term (inter-annual) time scale. The variability of correlation in space and time shows where and over what time-scale surface-depression storage processes could be hydrologically connected to runoff. The distinction of this time scale of correlation can then be further used to make inferences about the dominant hydrologic processes (e.g., overland flow or subsurface flow) that drive surface-depression storage contributions to runoff. Higher correlation is expected when there is a physical hydrologic connection between surface-depression storage and runoff; conversely, lower correlation is expected when surface-depression storage and runoff are hydrologically disconnected. Negative correlation values show inverse correlation between these variables. Correlations were calculated at monthly and annual time scale aggregations to measure differences in temporal resolutions of processes.

RESULTS

Results show the spatial distribution of the Spearman's rank correlation between NHM-PRMS fractional surface-depression storage and normalized runoff from the NHM-MWBM is different for monthly and annual values. The following sections further describe the temporal and spatial distribution of these correlation values.

Temporal Distribution of Correlation

The distribution of calculated monthly and annual Spearman's rank correlation coefficient values for each HRU with surface-depression storage for the CONUS ($n = 53,007$) is shown in Figure 5. There are more HRUs with a high annual Spearman's rank correlation relative to the monthly Spearman's rank correlation. To analyze these results further, the Spearman's rank correlation coefficient values were grouped into four categories to summarize the range of values for analysis: inversely correlated (from -1.0 to -0.2); no correlation (from >-0.2 to 0.2); correlated (from >0.2 to 0.75); highly correlated (from 0.75 to 1.0) (Table 2).

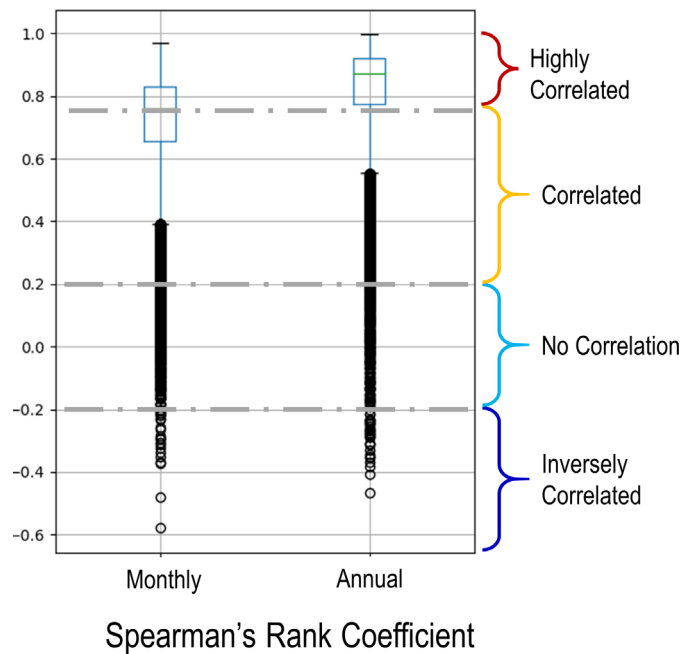


FIGURE 5. Boxplot showing the distribution of Spearman's rank coefficient between NHM-PRMS fractional surface-depression storage and normalized runoff from the NHM-MWBM for each HRU with surface-depression storage in the CONUS ($n = 53,007$). Distribution of values into categories is shown on the right axis, black circles represent single HRU correlation value outliers. Colors and brackets on the right show the values and assigned categories, consistent with the subsequent table and figure descriptions.

The number of HRUs within each category for monthly and annual Spearman's rank coefficient values is shown in Table 2. Overall, there are many more HRUs in both temporal resolutions that show correlation (23,825 Monthly, 11,016 Annual) and high correlation (28,279 Monthly, 41,655 Annual) than HRUs that show no (881 Monthly, 304 Annual) or inverse (22 Monthly, 32 Annual) correlation. There are fewer HRUs which show correlation at an annual analysis resolution ($n = 11,016$) than a monthly analysis resolution ($n = 23,825$). There are more HRUs that show high correlation at an annual analysis resolution ($n = 41,655$) than a monthly analysis resolution ($n = 28,279$). The dominant categories for both temporal aggregations are the "correlated" and "highly correlated" categories, which account for over 90% of HRUs for both temporal aggregations. The distributions of these does vary between the monthly and annual values. The monthly values are almost evenly distributed between "correlated" and "highly correlated"; whereas the annual values have much fewer in the "correlated" category and many more in the "highly correlated" category.

Several variables in addition to those listed in Table 1 were selected that describe the more geographically distributed nature of surface depressions

TABLE 2. The number (n) and percent of total HRUs ($n = 53,007$) within each category for monthly and annual Spearman's R coefficient values.

Category	Spearman's rank correlation range	Monthly		Annual	
		n HRU	% total	n HRU	% total
Highly correlated	>0.75 to 1.0	28,279	53.35	41,655	78.58
Correlated	>0.2 to 0.75	23,825	44.95	11,016	20.78
No correlation	>-0.2 to 0.2	881	1.66	304	0.57
Inversely correlated	>-1.0 to -0.2	22	0.04	32	0.06

(surface-depression storage density, number of depressions, surface-depression depth) from the NHM-PRMS simulation were grouped by monthly and annual Spearman's rank correlation categories to explore relative correlation. Of these, only seepage rate from the surface-depression storage and the surface-depression ET coefficient showed any difference

in distribution across Spearman's rank categories for monthly and annual groupings, shown in Figure 6. There is no substantial difference between any categories for the surface-depression storage seepage rate or ET coefficient. For the annual temporal dimension, there are differences in seepage rates and ET coefficients relative to the Spearman's rank category.

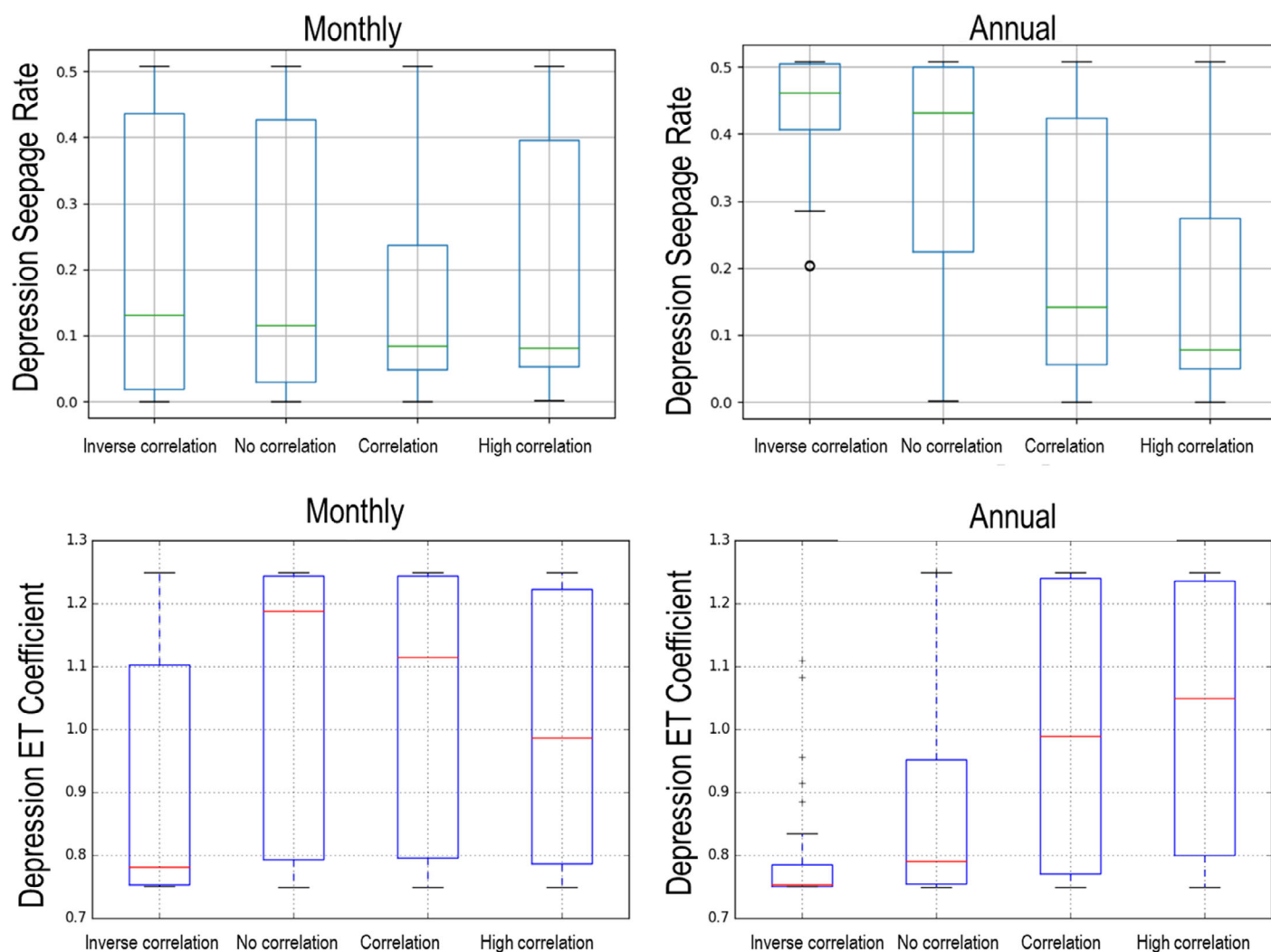


FIGURE 6. Boxplots by Spearman's rank category (defined in Figure 5: inverse correlation, no correlation, correlation, and high correlation) for monthly (left) and annual (right) values for surface-depression seepage rate (upper; in cm/day) and surface-depression evapotranspiration (ET) coefficient (lower).

Higher seepage rates are shown for HRUs with inverse connection, and the seepage rate decreases as the correlation becomes greater between surface-depression storage and runoff. There are lower ET coefficient values where there is inverse correlation, and the ET coefficient increases as the correlation becomes greater between surface-depression storage and runoff.

Spatial Distribution of Temporal Correlation

Correlation results by category (Table 2) are shown as choropleth maps for each HRU across the CONUS for monthly and annual data aggregations, which allows for visualization of the spatial variability of each temporal correlation. Maps of results show in Figures 6–8 show the geographic distribution of the HRUs included ($n = 53,007$) and not included (e.g., “No Data,” $n = 56,944$) in this analysis.

Monthly correlation between modeled fractional surface-depression storage and normalized runoff for HRUs are shown for the CONUS in Figure 7. This figure shows the majority of HRUs show a positive or strong positive correlation (yellow $n = 23,825$ and red $n = 28,279$, respectively) between monthly runoff and surface-depression storage. HRUs with inverse correlation (dark blue $n = 22$) are sparsely distributed in the deserts of California, Nevada and Utah. Areas with no correlation (light blue $n = 881$) include the basin and range areas of the west and eastern New York. East of the continental divide is mostly highly positively correlated, with a decrease in correlation in the Midwest. This figure shows that for the majority of the CONUS surface-depression storage is highly responsive to wetness conditions.

Annual correlation between modeled fractional surface-depression storage and normalized runoff for the HRUs with surface-depression storage is shown for the CONUS in Figure 8. Overall, the majority of

HRUs with depression storage show a positive or strong positive correlation (yellow $n = 11,016$ and red $n = 41,655$, respectively) between annual runoff and surface-depression storage. Areas of no correlation are in the Midwest and the montane west (light blue $n = 304$). There are not many HRUs with inverse correlation (dark blue $n = 32$), but those that show this statistic are located in the montane west.

To compare the monthly and annual correlation values, the difference between the categorical Spearman's rank correlations were calculated (Figure 9). This figure shows areas where correlation was higher for annual (blue) relative to areas where correlation was higher for monthly (red), where lighter colors show only one categorical difference ($>$) and darker colors show more than one categorical difference ($>>$) between Spearman's rank correlations. To generalize, this figure shows that interannual variability is more important than intra-annual or episodic variability for most of the country. This could be because inter-annual variability is larger, or that lags occur between wetness and surface storage accumulation that are missed by the monthly analysis. For areas such as the Midwest and Northeast that have higher annual than monthly correlation values this could be due to lake or other lag-inducing processes for runoff. HRUs with higher annual relative to monthly (annual $>$ monthly, blue) correlations are dominated by annual processes, whereas higher monthly relative to annual (monthly $>$ annual, red) correlations are dominated by monthly, shorter term processes.

The NHM-PRMS results were then aggregate into ecological regional groups for a broad-scale regional analysis to quantify if these ecological regional grouping would show any dominant modeled characteristics. For this analysis, groups of HRUs within the 86 level three ecoregions (U.S. Environmental Protection Agency 2013) of the CONUS were also explored for monthly and annual Spearman's R coefficient values, shown in Figure 10. Overall, most ecoregions show

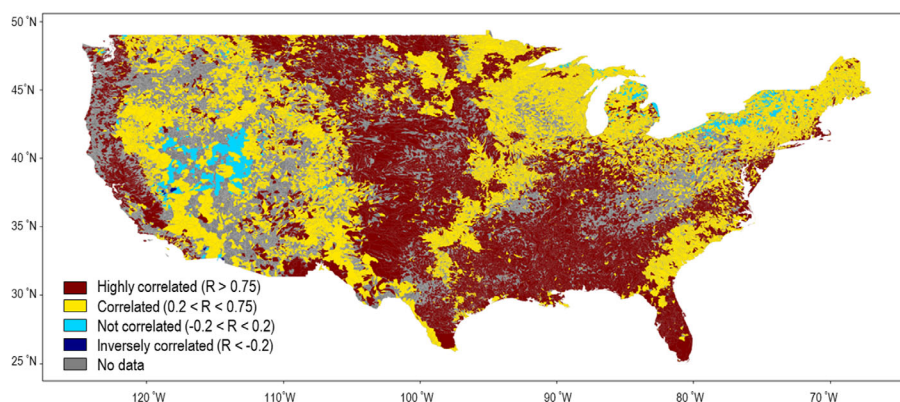


FIGURE 7. Spatial distribution of Spearman's rank correlation between monthly runoff and surface-depression storage.

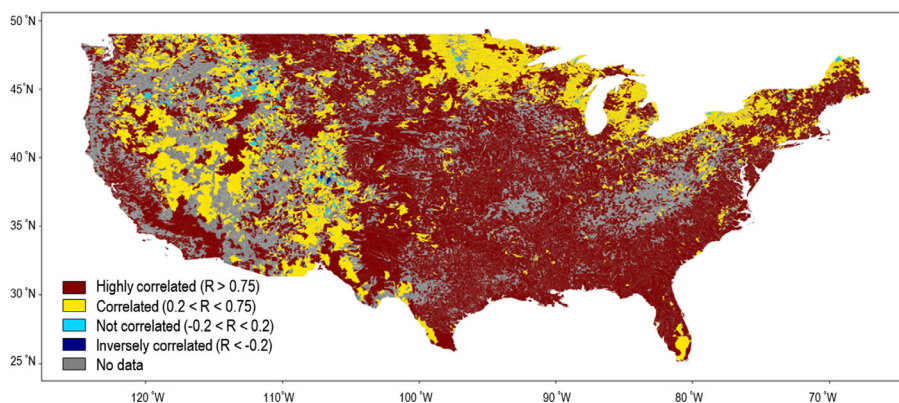


FIGURE 8. Spatial distribution of Spearman's rank correlation between annual surface-depression storage and runoff.

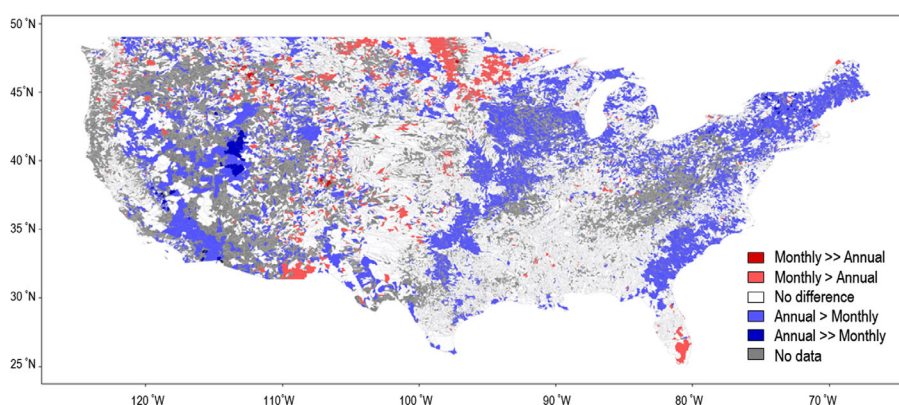


FIGURE 9. Difference between monthly (Figure 7) and annual (Figure 8) Spearman's rank correlation coefficient values for surface-depression storage and runoff for each HRU over the CONUS.

ranges of annual Spearman's rank coefficient values greater than ranges of monthly Spearman's rank coefficient values. There are five ecoregions which show higher monthly correlation relative to annual correlation: Cascades, Coast Range, Lake Agassiz Plain, Northern Glaciated Plains and Southern Florida Plain. These areas are consistent with results shown in Figure 8, apart from instances at the HRU-scale in the Northern, Middle, and Southern Rockies regions. It is likely that the range of values in these interior montane environments is too wide for there to be an ecoregion-wide trend in the distribution. These five ecoregions with higher monthly Spearman's R coefficient values are more likely to be good candidates for the proxy calibration strategy developed in the Prairie Pothole Region (Hay et al. 2018).

DISCUSSION

This study advances how landscape-scale surface depressions are simulated in national-extent models.

This study examines the spatial variability of the temporal correlation between modeled fractional surface-depression storage and normalized runoff for the CONUS. Numerous studies show the importance of hydrologic connectivity of wetlands and surface storage dynamics on ecosystems (Leibowitz 2003; Vanderhoof et al. 2016), with very limited analyses at a CONUS-or global extent (Cohen et al. 2016; Pekel et al. 2016). Description of complex hydrologic processes such as surface-depression storage in both a physically realistic yet computationally efficient way is an ongoing challenge. The correlation between fractional surface-depression storage and normalized monthly runoff identifies where monthly normalized runoff volume might be used as a proxy for change in surface-depression storage for model calibration of surface-depression storage processes.

Study results show the temporal and spatial variability of correlation between modeled fractional surface-depression storage and normalized runoff by HRU for the CONUS. Previous studies on residence times of water through the landscape have used a distribution of transit times or mean transit times to describe the movement of water from precipitation to streamflow to

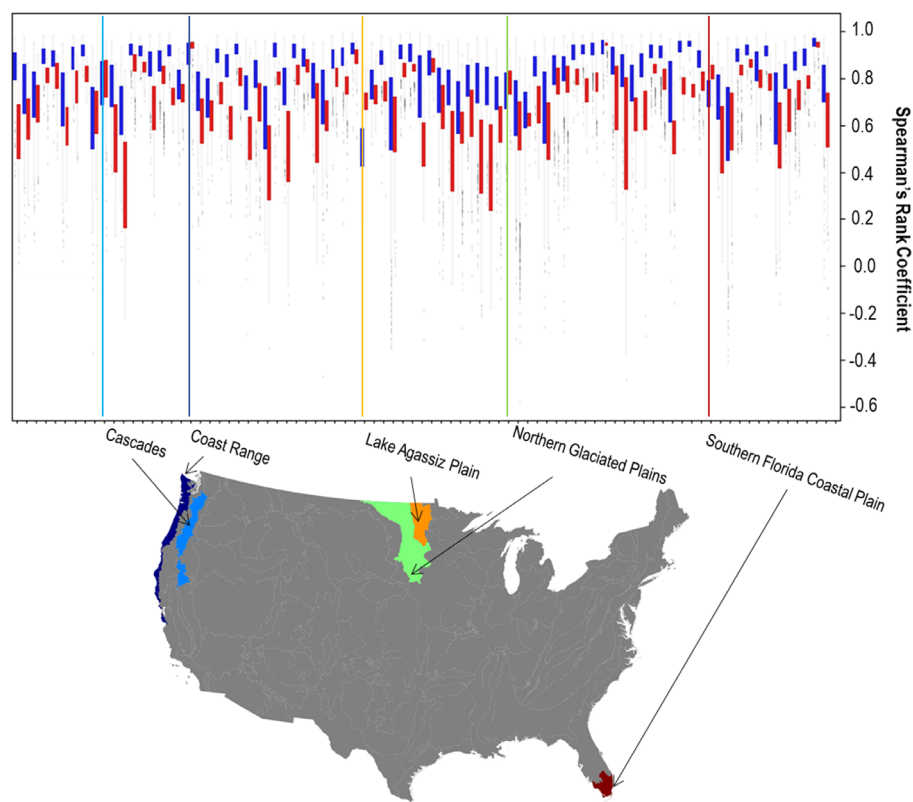


FIGURE 10. Ranges of Spearman's rank correlation coefficient values for annual (blue) and monthly (red) surface-depression storage and runoff for all HRUs within the 86 level III ecoregions of the CONUS (alphabetically from left to right). Ecoregions with annual ranges higher than monthly ranges are labeled and shown in color on the map, with corresponding colored lines to indicate which boxplots relate to these regions.

describe storage processes either at a fine spatial resolution (e.g., catchment-scale) (McGuire and McDonnell 2006; Soulsby et al. 2011; Heidebüchel et al. 2012) or broad temporal resolution (e.g., geologic time scale) (Maxwell et al. 2016). This investigation was limited by the monthly aggregation of daily model values; investigation on submonthly and time scales for the CONUS would further improve our understanding of these processes at a finer temporal resolution. The results of this study provide analysis at a broader spatial resolution and limited temporal resolution, which is applicable to estimation of annual water budgets at a national extent.

Results show that overall annual correlation is more extensive than monthly correlation over the CONUS. The geographic areas where monthly correlation is greater than annual correlation are generally dominated by seasonal, short-term, "dynamic" storage. Examples include the desert southwest and higher elevation montane regions where high-intensity seasonal precipitation events (e.g., convective storms) or snow-melt are dominant processes in the water budget. These short-term, high-magnitude fluxes of water into the system likely drive filling and spilling of surface depressions, producing higher short-term (monthly)

correlation. Conversely, areas where annual correlation is greater than monthly are likely dominated by annual storage, where there is a longer lag time between spill and runoff, following the general temporal trends of an annual water balance.

The relation of model variables including surface-depression seepage rate and surface-depression storage ET to the categorical Spearman's rank correlation coefficient values were investigated. At the annual time scale, the high seepage rate HRUs show an inverse correlation between modeled surface-depression storage and runoff, which could mean decreases in wetness are due to more water from storage seeping to groundwater. Increases in wetness resulting in an increase in both surface-depression storage and runoff could mean less water seeping into the ground, shown as high correlation where seepage rate is low.

In some cases where inverse correlation occurs there are higher seepage rates, perhaps indicating the lack of hydrologic connection between groundwater and surface-water tables. Conversely, in areas with lower seepage rates there could be higher hydrologic connection because of greater spillage (overland flow and interflow) leading to runoff. This trend may indicate that the correlation of surface-depression storage

and runoff is because of spillage rather than contribution from groundwater. Conversely, model ET coefficient values are shown to increase with increased correlation between surface-depression storage and runoff. Higher ET coefficient values indicate greater water loss to ET when there is more correlation between surface-depression storage and runoff. This relation may indicate more water available in the system that is lost to ET at times when there is more correlation between surface-depression storage and runoff. These results may be in part because of the specific parameterization of surface-depression storage in the PRMS; all surface depressions are set to “open,” allowing them all to spill. If the surface-depression storage settings allowed “closed” surface depressions, no spillage would be allowed, and correlation between runoff and surface-depression storage might then be attributed to groundwater contribution to runoff.

While the NHM-PRMS CONUS application shows significant advancement for representation of surface-depression storage processes at this scale, there is room for improvement. These improvements could include (1) refined spatial representation (surface-depression geometry, dynamic surface-water extent), and characterization to allow for use of both “open” and “closed” depressions, and (2) groundwater contribution into surface-depression storage. With respect to spatial representation, the first improvement could be utilizing the ability to define the geometry of the surface depressions. The default setting, which was used for this study, is for all surface depressions to be rectangular. Specification of other geometries, such as bowl or triangular shapes would change ET by more accurately representing the portion of a given volume of water is exposed to open-water evaporation. This may then allow for a more accurate representation of parameter `dprst_et_coef` (Table 1). In addition to this aspect of spatial representation, the expansion and contraction of these geographically distributed surface depressions have been shown to influence surface-water connectivity (Vanderhoof et al. 2016). CONUS (Jones 2019) and global (Pekel et al. 2016) scale characterization of time-series surface-water extent (and changes in surface-water elevation (Hay et al. 2018) using remote sensing data products could also help improve model representation of surface-depression storage processes leading to better understanding of hydrologic connectivity in these complex systems.

CONCLUSIONS

This study explores the correlation of simulated monthly runoff and daily aggregated surface-depression

storage processes in CONUS-extent models. The ability to run two models with different temporal resolutions (daily PRMS and monthly MWBM) on the same spatial discretization is novel and was integral to the ability to perform these analyses. Monthly and annual aggregated fractional surface-depression storage from the daily hydrologic simulation model, the USGS PRMS, and monthly normalized runoff from a monthly model, the USGS MWBM, were compared using Spearman's rank correlation test for each HRU of the NHM infrastructure. Results show the utility of CONUS-extent, deterministic models in investigation of individual hydrologic processes, such as surface-depression storage. Results from this analysis show the spatial extent of where monthly normalized runoff volume may be used as a proxy for surface-depression storage for a daily model calibration. The monthly and annual correlation values provided a quantitative measure to determine the which of these temporal resolutions are most relevant for the analysis of surface-depression storage processes for each HRU. The spatial distribution of monthly and annual correlations shows where monthly dynamic and long-term storage are more common, respectively. This analysis shows higher correlation value for annual time scales are generally more common for the CONUS; however, higher correlation values for monthly time scales are more frequent in focused areas where surface-depression storage areas are more prevalent, such as the Prairie Pothole Region.

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